

plays an important role in identifying the network and host portion of an IP address. We shall come to the subnet mask gradually as we proceed with understanding the classes of IP addresses.

7.1.2 IP Classes

IP addresses are classified under three classes, namely Class A, Class B and Class C.

Class A ranges from 0.0.0.0 to 126.255.255.(1 to 254). The subnet mask for this class is 255.0.0.0. This will mask the first number in your IP address as the network address and the remaining becomes your host address. Hence the total number of networks is 127, and the number of hosts per network is $256 \times 256 \times (256-2)$. In this entire range, 10.x.x.x is reserved as private IPs, and 127.0.0.0 is reserved as loopback localhost.

Class B IP address ranges from 128.0.0.0 to 191.255.255.(1 to 254). The subnet mask for this class is 255.255.0.0. Hence the first two bytes of an IP address is masked as the network portion. IP address ranging from 172.(16 to 32).x.x are reserved as the private range.

Class C has the minimum number of hosts amongst all the classes. Class D and E do not fall under the normal users category—they are reserved for Multicast and Military/R&D respectively. Class C continues after the Class B from 192.0.0.0 to 223.255.255.(1 to 254). The subnet mask is 255.255.255.0. The private IP range falls within 192.168.0.x.

It is not obligatory to use the prescribed subnet mask. For instance, some ISPs use Class A IPs, say 10.15.10.28, with a 255.255.255.0 subnet. This is possible, but it restricts a host to network of 10.15.10.x, whereas it could have actually belonged to a wider network of 10.x.x.x had the subnet been 255.0.0.0. In some Linux systems, you will notice an IP address given as 192.168.0.12/24. This is a combination of an IP and a subnet mask. 255.0.0.0 is denoted as /8, while 255.255.0.0 and 255.255.255.0 are denoted as /16 and /24 respectively.